Ghislain THEUBO

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LinkedIn | Bilingual English/French (B2+) | GitHub

Professional Summary

Application Development Engineer (Ensimag) with strong **Python** and **C/C++**, production scientific/data software at **Gustave Roussy & CentraleSupélec MICS**, real-time sensor pipelines at **INRIA** (C/FreeRTOS), and industrial software craft at **Dassault Systèmes**. Practiced in Agile, unit/functional tests, Git, CI/CD, databases, and Linux. Eager to apply software engineering to space-plasma data tools (analysis, intercalibration, scientific products, databases, diffusion) for **Comet Interceptor / COMPLIMENT** at LPC2E – CNRS (Orléans, CDI de mission 84 months, start from Jan 2027).

Core Strengths

Python · **C/C++** · Scientific / sensor data pipelines · Algorithms & databases · Agile / TDD / CI/DevOps · Linux & Windows · English B2+ · Teamwork with scientists & instrument engineers · Autonomy & rigor

Experience

Shift Technology – Platform Software Engineering — Paris, France
Software Engineer (CDI – Platform / Backend) — *Sept 2026 – Nov 2026*

- Contributed to the **platform / foundations** squad building shared services used by product teams: design and implementation of **C# / .NET** components and **high-performance APIs** on **Microsoft Azure**.
- Worked with cloud data and messaging building blocks (**Azure Cosmos DB**, **Service Bus**), configuration/transversal services, and infrastructure-as-code awareness (**Terraform**) in a hybrid SaaS environment.
- Practiced production engineering habits: design discussions, code reviews, monitoring mindset, incremental delivery, and collaboration with platform and product engineers.

Gustave Roussy & CentraleSupélec MICS Lab — Paris / Saclay, France
Software Engineer (CDD – Backend / Scientific Data Platforms) — *Jan 2026 – Sept 2026*

- Built and maintained production software for **research / clinical scientific workflows**: REST APIs in **Python/FastAPI**, validation, RBAC, structured logging, observability.
- Modeled and tuned **PostgreSQL** (indexing, query profiling — **∼30%** faster critical reads); ensured coherent, auditable data products for researchers.
- Docker + **GitHub Actions** CI; technical documentation for data flows and operational handoff to scientific users.
- Daily collaboration with domain experts — translating measurement/analysis needs into robust, testable tools (same scientist–engineer loop as LPC2E).

Dassault Systèmes – 3DEXPERIENCE / 3DSwym R&D — Vélizy-Villacoublay, France
Software Engineer Intern (Full-Stack Module Rewrite) — *Mar 2025 – Sept 2025*

- Full overhaul of a legacy module with **Java 17 / Spring Boot / Hibernate**; applied **TDD** (JUnit), design patterns, Agile/MoSCoW with product Champions.
- Results: **∼35%** performance improvement, **∼85%** test coverage — quality, versioning, and maintainability habits aligned with CNRS software engineering expectations.

INRIA Grenoble – Humanlab / Froll'N'Roll — Montbonnot-Saint-Martin, France
Research Intern – Real-Time Sensor / Control Software — *May 2024 – Aug 2024*

- Developed **C/FreeRTOS** and Python modules for sensor-driven control on constrained hardware; redesigned pipelines (**∼40%** responsiveness).
- Worked with instrumentalists/scientists; documented codebases; continuous validation against real measurements — transferable to electrical-sensor data tools for space plasma.

LIG – Laboratoire d'Informatique de Grenoble — Grenoble, France
Cloud Infrastructure Intern (OpenStack / Grid'5000) — *Feb 2025 – Mar 2025*

- Automated experiments and performance analyses under Linux research infrastructure — reproducible tooling mindset.

Selected Projects (mission-aligned)

Plasma-like Sensor Pipeline Lab | Python, NumPy/SciPy, C++ extension notions, HDF5/CSV — GitHub

- End-to-end pipeline: ingest multi-sensor time series → cleaning/intercalibration stubs → derived scientific products → indexed store + export for analysts.
- Unit/functional tests, CI, Linux-first; documented product levels (L0/L1-style) for diffusion to a mock data centre.

Scientific API + DB for Research Data | Python/FastAPI, PostgreSQL — Gustave Roussy / personal

- Production APIs and schemas that make research datasets queryable, coherent, and handoff-ready — same class of work as COMPLIMENT data generation.

C++ Numerical Kernels + Python Bindings Sketch | C++, pybind11 notions, Catch2 — GitHub

- Performance-minded compute kernels callable from Python — ramp for mixed Python/Fortran/C++ scientific stacks at LPC2E.

Technical Skills

Languages: Python (strong), C/C++ (strong), Java/Spring; Fortran ramp-up readiness
Software eng.: Agile, unit & functional tests, Git, CI/CD, DevOps notions, quality discipline
Data / science: algorithms, databases (SQL), scientific computing (NumPy/SciPy), sensor pipelines
OS: Linux (daily), Windows
Soft: project ownership, written/oral communication, international English B2+, scientist collaboration

Education & Certifications

Grenoble INP - Ensimag Grenoble, France
Engineering Degree – Information Systems / Software Engineering (Double Degree) *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé Yaoundé, Cameroon
Master's Degree in Computer Engineering – Top 12% of class *2019 – 2023*

Certifications: Programming Foundations – TDD (LinkedIn, 2025) · Primerli – Insurance Industry Foundations (2025)

Languages & Extra

English: bilingual / professional (B2+) · **French:** fluent · **Spanish:** intermediate
Extra: tutoring · tennis, football · motivation for space research software (Comet Interceptor)